

Molecular Surveillance for Lymphoproliferative Disease Virus and Reticuloendotheliosis Virus in Rio Grande Wild Turkeys (*Meleagris gallopavo intermedia*) in Texas, USA

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ABSTRACT: Reticuloendotheliosis virus (REV) and lymphoproliferative disease virus (LPDV) are avian retroviruses that can cause neoplastic disease and present with similar pathologies. Lymphoproliferative disease virus has been reported in the Eastern US and states bordering Texas, USA, but has not been previously detected within the state. In a prior study, we detected REV in native Rio Grande Wild Turkeys (*Meleagris gallopavo intermedia*) and an Eastern Wild Turkey (*Meleagris gallopavo silvestris*) originating from West Virginia. Given LPDV detection in states bordering Texas and our finding of an REV-positive Eastern Wild Turkey imported from a LPDV endemic region, we sought to determine LPDV prevalence in Texas and continue surveillance for REV. During 2018–20, dried blood spots from 373 individual Rio Grande Wild Turkeys from 20 different counties were tested for the presence of proviral REV or LPDV DNA. In affected counties, approximately 4% of individuals were infected with REV (7/197) or LPDV (10/273) and one bird was coinfecting with both viruses. Phylogenetic analysis indicated a close relationship of the LPDV isolates to variants from other Southern and Central states. This study provides molecular evidence of LPDV in Texas, and continued surveillance is necessary to determine the potential effects of the virus on reproductive success, coinfections, and overall health of Wild Turkey populations.

Key words: Avian retrovirus, lymphoproliferative disease virus, reticuloendotheliosis virus, Wild Turkeys.

Reticuloendotheliosis virus (REV) and lymphoproliferative disease virus (LPDV) are avian retroviruses associated with neoplastic disease (Fadly 1997). Reticuloendotheliosis virus is an oncogenic gammaretrovirus that infects B-lymphocytes of Galliformes, Anseriformes, and Passeriformes and can cause immunosuppression (Nair et al. 2013; Ferro et al. 2017). Clinical abnormalities associated

with REV infection include anemia, non-neoplastic runting, lymphoma, and feathering abnormalities (Nair et al. 2013). Lymphoproliferative disease virus, an alpharetrovirus, infects Wild Turkeys (*Meleagris gallopavo*) and is associated with lymphoid tumors (Biggs et al. 1978). Reticuloendotheliosis virus and LPDV are rarely associated with clinical disease. However, retrovirus infections are associated with decreased reproductive success, hatchability, and sexual maturation in domestic poultry (Payne 1998; Wei et al. 2012). First identified in the UK in 1972, LPDV has been credited with neoplastic disease outbreaks in Europe and Israel (Biggs et al. 1978; Ianconescu et al. 1983). Lymphoproliferative disease virus proviral DNA was first detected in the US in 2009 in a Wild Turkey in Arkansas and has been detected in Wild Turkeys from 24 US states, including Oklahoma (prevalence 26%; 7/27), Louisiana (57%; 57/96), and West Virginia (55%; 26/47; Allison et al. 2014; Thomas et al. 2015; Alger et al. 2017). In a 2016–17 survey, we identified five REV-positive Rio Grande Wild Turkeys (*Meleagris gallopavo intermedia*) and one REV-positive Eastern Wild Turkey (*Meleagris gallopavo silvestris*) imported from West Virginia (Stewart et al. 2019). Given LPDV detection in states bordering Texas and this finding of an REV-positive imported Eastern Wild Turkey, our objectives were to determine REV and LPDV distribution and prevalence in Texas, USA, continue surveillance for REV, and add to our understanding of LPDV host range within a subspecies of Wild Turkeys.

Outwardly healthy Rio Grande Wild Turkeys (adult=272, juvenile=63; female=202, male=133; unknown age or sex=38) were captured from 20 different counties of Texas, with walk-in funnel traps, drop nets, or rocket nets by Texas Parks and Wildlife Department (TPWD) biologists. Capture was authorized by an Employee Scientific Collection Authorization Permit as part of the TPWD Eastern Wild Turkey restoration program. Dried blood spots were collected on sampling cards constructed from Whatman grade 3MM paper (GE Healthcare, Chicago Illinois, USA) prelabeled with a unique identification number. Sterile lancets were used to puncture the wing vein and blood was spotted on the cards and allowed to dry for approximately 1 h. From each dried blood spot, 5-mm squares were cut, mixed with 75- μ L HotSHOT alkaline lysis reagent, and incubated for 30 min at 95 C while shaking, then cooled on ice for 5 min before the addition of HotSHOT neutralization buffer (Truett et al. 2000). Extracted genomic DNA was stored at -20 C until initial screening by quantitative PCR (qPCR) assay for REV *env* and LPDV *env* genes to test for viral infection and the pan-avian *GAPDH* gene, which served as a DNA extraction control. Reaction conditions were optimized (see Supplementary Methods 1). Assembled 20- μ L reactions contained 500 nM forward and reverse primer and 200 nM probe for each target, 10 μ L PrimeTime® Gene Expression 2X MasterMix (Integrated DNA Technologies, Coralville, Iowa, USA), 5 μ L nuclease-free water, and 3 μ L DNA in each well. Standard PCR (see Supplementary Methods 2) was used to verify samples that were REV- or LPDV-positive by qPCR and was assembled in 25- μ L reactions using 5.5 μ L nuclease-free water, 12.5 μ L OneTaq Hot Start 2X MasterMix (New England Biolabs, Ipswich, Massachusetts, USA), 2 μ L of the eluted DNA, and 1 μ L of the forward and reverse primers (200–400 nM final concentration) targeting either the REV 3' long terminal repeat, LPDV p31/CA, or pan-avian *GAPDH* (Aly et al. 1993; Allison et al. 2014; Olias et al. 2014). Detectable PCR products were submitted to Texas A&M–Corpus Chris-

ti Genomics Core Laboratory (Corpus Christi, Texas, USA) for Sanger sequencing (GenBank accession no. OL960637–OL960653) using the same forward and reverse primers. The similarity in nucleotide sequence identity to published REV and LPDV proviral sequences was determined by BLASTn (Altschul et al. 1990).

Evolutionary analyses were conducted in MEGA X (version 11.0.11) using the Maximum Likelihood method based on the General Time Reversible model (Nei and Kumar 2000; Kumar et al. 2018). Initial trees for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Maximum Composite Likelihood approach and then selecting the topology with a superior log-likelihood value. A discrete gamma distribution was used to model evolutionary rate differences among sites (5 categories [+G, parameter=0.4955]). The initial analysis involved 195 nucleotide sequences previously published from the US and Israel (Allison et al. 2014; Thomas et al. 2015). Redundant sequences were removed from the dataset and the remaining sequences were re-analyzed. The final analysis involved 98 nucleotide sequences, with a total of 423 positions in the final dataset.

We identified seven REV-positive and 10 LPDV-positive birds (see Supplementary Figs. S1 and S2). In affected counties, 7/197 (4%) of individuals were infected with REV and 10/273 (4%) with LPDV (Table 1). Two infected Wild Turkeys, one REV-positive and the other LPDV-positive, were identified in Cottle County within the Rolling Plains ecoregion in North Texas. In the Edwards Plateau ecoregion in Central Texas, we identified four REV-infected and eight LPDV-positive Wild Turkeys, as well as one individual that was co-infected. Within South Texas, we identified one REV-positive Wild Turkey in Live Oak County. Of the REV-positive Wild Turkeys, 5/7 (71%) were adults, as were 9/10 (90%) of those infected with LPDV. The sex ratio (female:male) was 5:2 for REV-infected birds and 4:1 for LPDV-infected. Phylogenetic analysis revealed that most

TABLE 1. Reticuloendotheliosis virus (REV) and lymphoproliferative disease virus (LPDV) prevalence in Rio Grande wild turkeys (*Meleagris gallopavo intermedia*; n=373) in Texas, USA, by county. Percent positive and frequency was determined by the number of REV- and LPDV- positive samples divided by the total number of samples.

County	REV		LPDV	
	Percent positive	Frequency	Percent positive	Frequency
Bell	0	0/11	0	0/11
Comal	0	0/2	50	1/2
Cottle	1	1/97	1	1/97
Gillespie	0	0/43	2.3	1/43
Karnes	0	0/4	0	0/4
Kendall	10	1/10	20	2/10
Kerr	0	0/36	2.8	1/36
La Salle	0	0/5	0	0/5
Lampasas	0	0/19	0	0/19
Live Oak	20	1/5	0	0/5
Llano	0	0/2	0	0/2
Mason	0	0/19	0	0/19
McMullen	0	0/1	0	0/1
Menard	0	0/8	0	0/8
Navarro	0	0/3	0	0/3
Palo Pinto	0	0/6	0	0/6
Schleicher	4.7	4/85	4.7	4/85
Shackelford	0	0/5	0	0/5
Williamson	0	0/3	0	0/3
Wise	0	0/8	0	0/8
Unknown	0	0/1	0	0/1

(7/10) LPDV p31/Ca sequences from Texas grouped with sequences from other Southern and Central states (Fig. 1), suggesting a potential origin for LPDV strains from Texas. Three sequences grouped separately from previously published US sequences, while all were distinct from the Israeli strain.

This study, our previous survey in 2016–17, and a 2002 report indicate that REV is circulating within Rio Grande Wild Turkey populations in Texas (Peterson et al. 2002; Stewart et al. 2019). Our current study reports the largest detected geographic range of REV in Wild Turkeys within Texas. In 2002, REV-positive Wild Turkeys were identified in the Edwards Plateau ecoregion (Peterson et al. 2002). During 2016–17, we identified addi-

tional REV-positive Wild Turkeys in the Edwards Plateau and Rolling Plains ecoregions (Stewart et al. 2019). In this study, we detected REV-positive Wild Turkeys not only in these previously tested ecoregions, but also in the South Texas Brush Country where turkeys had not previously been tested for REV (Ferro et al. 2017).

Our finding that LPDV is present in Texas Wild Turkeys supports previous studies indicating that LPDV is widespread across the Eastern and Central US and that Central states have lower prevalence than the Northeastern, Mid-Atlantic, and Southeastern states (Thomas et al. 2015; Alger et al. 2017). Wild Turkey susceptibility to LPDV may vary by subspecies (McDougall et al. 1978). Prevalence of LPDV correlates with Wild Turkey subspecies, and Rio Grande Wild Turkeys are the predominant subspecies in Texas, Oklahoma, and Kansas as compared to the Eastern Wild Turkey, which is most common in the Northeastern, Mid-Atlantic, and Southeastern US. A limitation of our study was the use of dried blood spots, which may decrease assay sensitivity as compared to whole blood samples (Smit et al. 2014). We found that adult female Wild Turkeys had a higher proportion of LPDV and REV infections than did juveniles or males. Other reports suggested that LPDV is more likely to be found in adult birds and is fatal to young (4–16 wk) birds. However, infection in young poultry may be more prevalent than adult surveys suggest, as a result of increased mortality and scavenging of carcasses (Biggs et al. 1978; McDougall et al. 1978; Thomas et al. 2015; Niedringhaus et al. 2019). Although we detected the majority of LPDV infections in females, evidence of sex as a predictor of infection varies (Alger et al. 2017; Niedringhaus et al. 2019). While subclinical infections with LPDV may not pose an immediate threat, the potential effects on reproductive success, coinfections, and the overall health of Wild Turkey populations warrant further surveillance.

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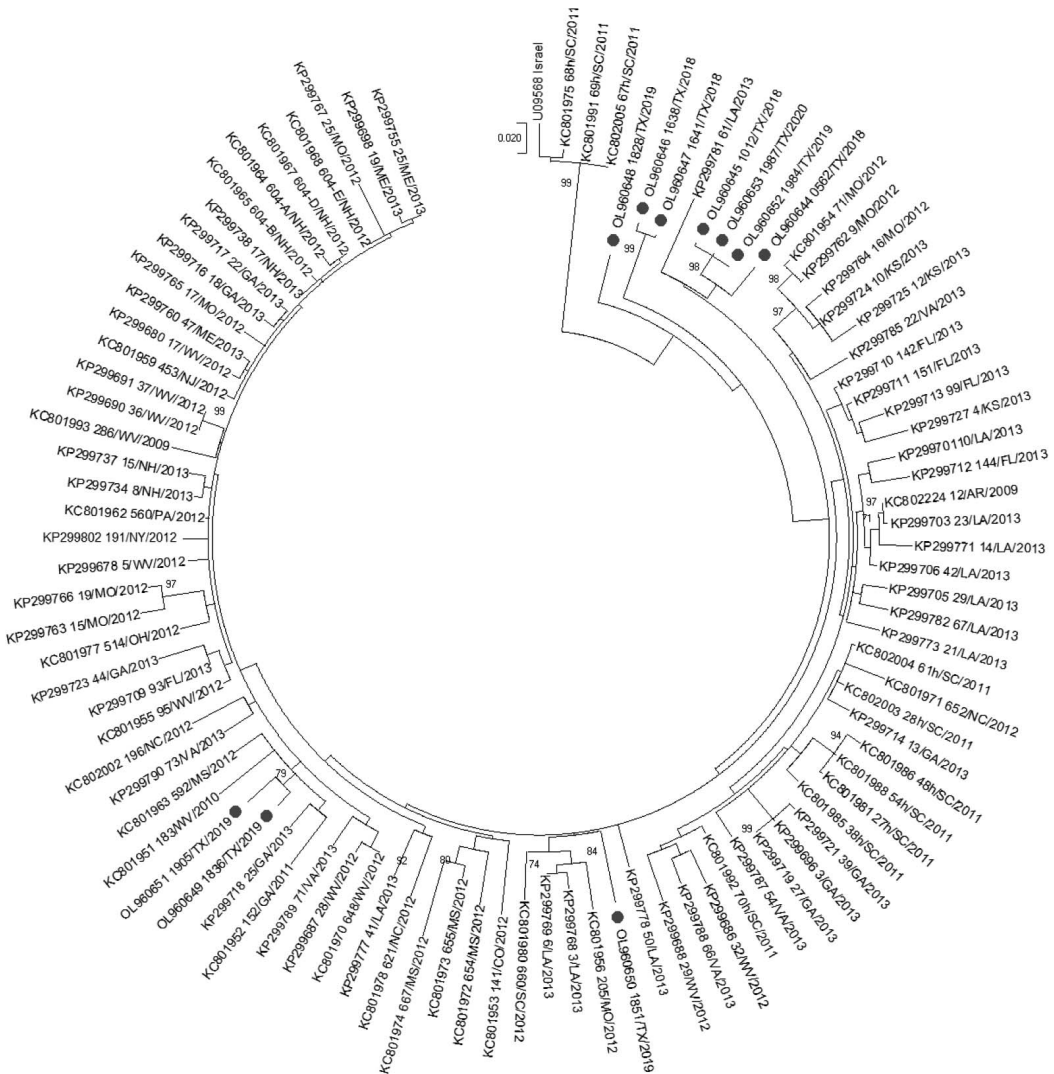


FIGURE 1. Phylogenetic analysis of LPDV p31/Ca sequences isolated from 10 Rio Grande Wild Turkeys (*Meleagris gallopavo intermedia*) in Texas, USA, through 2018–20. The prototype Israeli strain and 84 isolates from the US are included. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. The tree with the highest log likelihood (−4236.65) is shown and circles denote the sequences isolated in Texas.

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SUPPLEMENTARY MATERIAL

Supplementary material for this article is online at <http://dx.doi.org/10.7589/IWD-D-22-00023>.

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